

Open source data governance tools include Amundsen, DataHub, Apache Atlas, Magda, Open Meta Data, Egeria, and TrueDat. These tools offer features like metadata management, and data cataloguing and collaboration.

Data quality: It helps in ensuring the accuracy, completeness, and consistency of data. Data quality management involves identifying and correcting errors, standardising formats, and maintaining a high level of data integrity.

The best open source data quality tools include Cucumber, Deequ, dbt Core, MobyDQ, Great Expectations, and Soda Core. These tools help in the automation of data validation processes, data cleaning and monitoring.

Data stewardship: This is about assigning roles and responsibilities related to data management. Data stewards are designated individuals or teams entrusted with overseeing the appropriate use, integrity, and secure storage of enterprise data. They serve as a vital bridge between IT and business units, ensuring that data conforms to the enterprise's established quality and consistency standards. Key responsibilities include defining and standardising data elements, monitoring data quality, and

collaborating with IT to resolve any technical challenges.

Data policies and procedures:

These establish and enforce policies for how data is collected, stored, shared, and used. An enterprise's central data management policies prescribe permitted and prohibited practices at every stage of the data life cycle. They ensure compliance with internal standards and external regulations.

Data security: Establishing proper security protocols helps in reducing the risk of data breaches and threats. It also safeguards sensitive information. Encryption, access controls, authentication, and intrusion detection systems help in protecting data across the life cycle.

Open source data security tools that are widely used include Metasploit, OSSEC, OpenVAS, Snort, KeePass, and ClamAV. These tools can be integrated into various security strategies to protect against a wide range of cyber threats.

Metadata management: This helps in keeping track of data definitions, relationships, and structures. It's essentially data about data. Metadata functions as the contextual glue that transforms isolated data points into

coherent, actionable assets. It captures essential attributes covering:

- Creation timestamp
- Authorship and ownership of data
- Source provenance
- Relationships with other data elements

Leading open source metadata management tools are Apache Atlas, Amundsen, LinkedIn DataHub, Metacat Data Catalogue, Open Metadata, and Marquez.

Master data management:

Master data management (MDM) is a process for ensuring the accuracy, consistency, and completeness of critical data elements, such as customer data and product data. Master data is standardised, matched, merged, enriched, and validated according to governance rules.

Key open source players in the MDM area are Talend Open Studio for MDM, AtroCore, and Pimcore.

Data storage: This helps in determining where and how data will be stored as an enterprise data repository. It covers both structured and unstructured data sources, which include databases, data warehouses and data lakes.

Popular open source tools in the data storage area are Hadoop, LakeFS, Cassandra, and Neo4J. These tools provide scalability, robustness and performance in managing and analysing large datasets in various applications.

Data privacy and

compliance: This ensures adherence to regulations and ethical considerations. Privacy implements controls to prevent unauthorised access.

Regulatory frameworks such as the European Union's General Data Protection Regulation (GDPR) and California's Consumer Privacy Act (CCPA) impose stringent requirements on how businesses collect, process, and safeguard personal data.

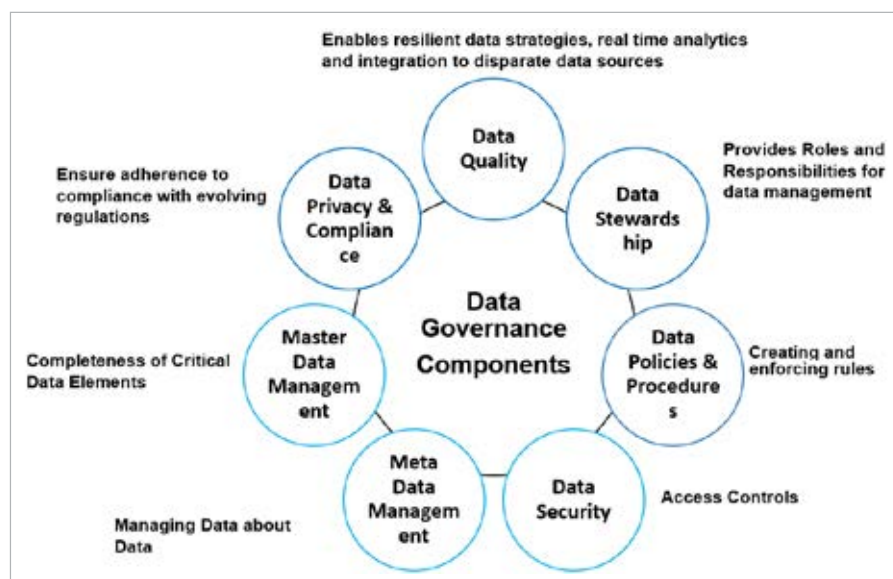


Figure 1: Key components of data governance